HARDWARE BASED IR REMOTE RELAY CONTROLLED SYSTEM

Embedded Systems | Digital Logic (No Microcontroller) | Analog Signal Conditioning

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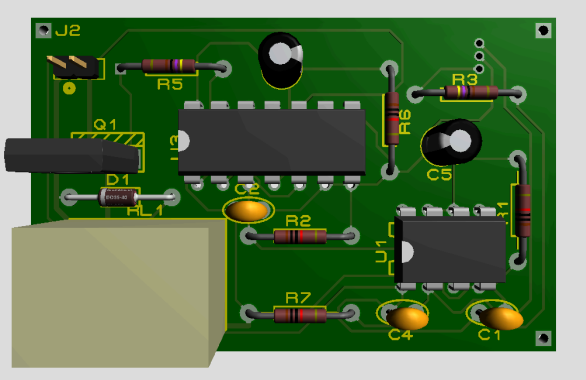
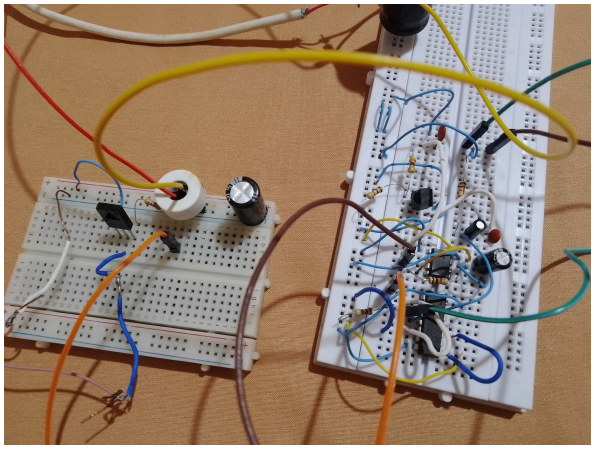
Overview

this project implements an IR remote-controlled relay toggle system using discrete hardware components only. The design processes IR receiver output through signal conditioning, pulse shaping, and flip-flop logic to control a relay-driven load. The system was developed to explore hardware-based digital control .

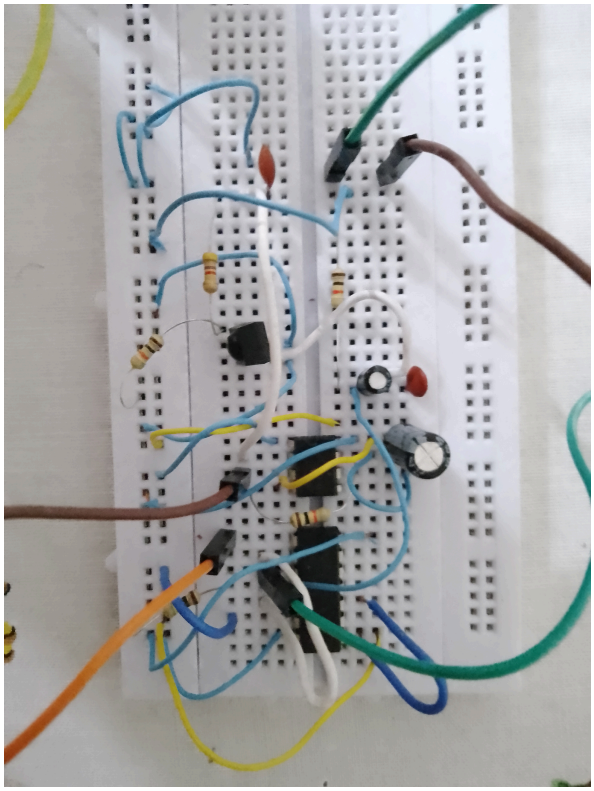


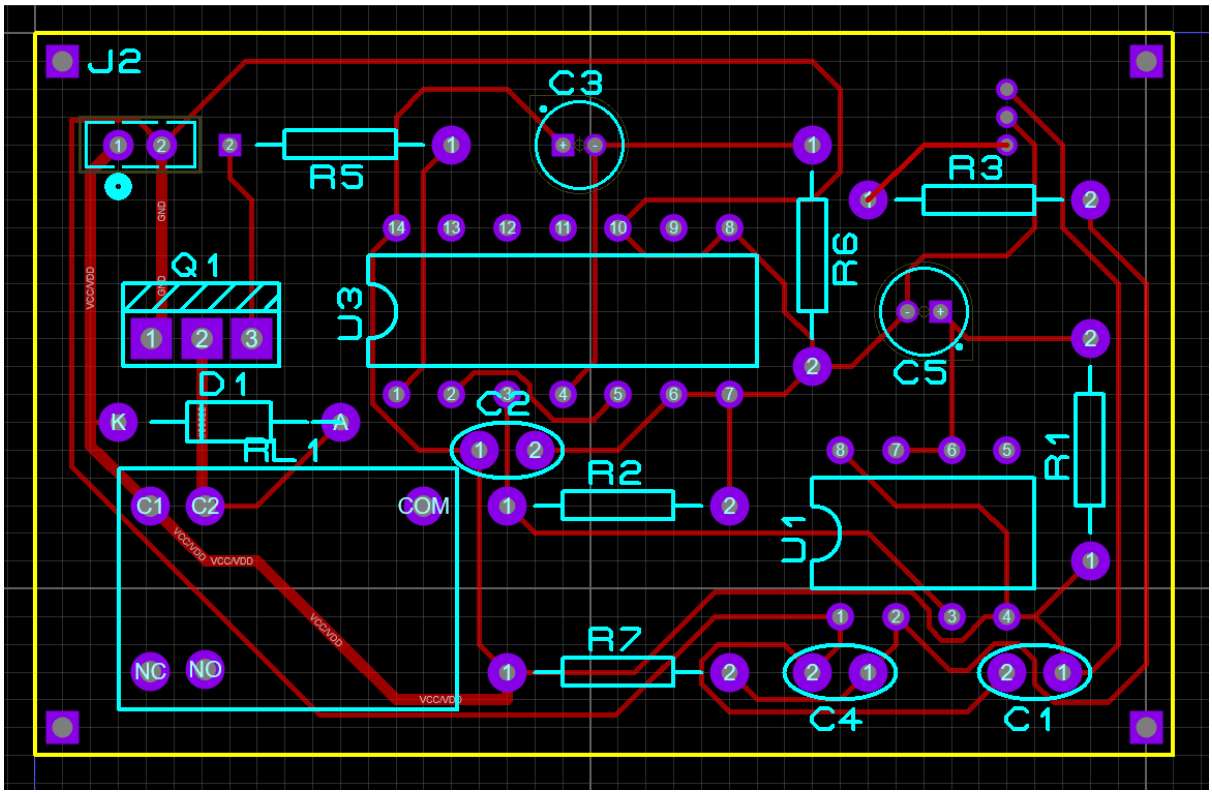
Proteus Ares PcB design



working breadboard version

click here for video demonstration



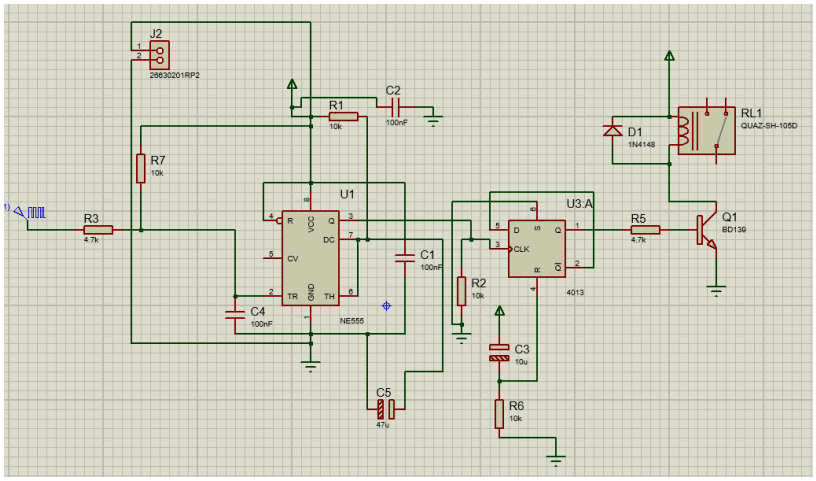

J2
C3
R5
R3
R6
Q1
U3
C5
D1
K
A
RL1
C2
R2
R1
C1
C2
COM
U1
NC
NO
R7
C4
C1
VCC/VDD
GND

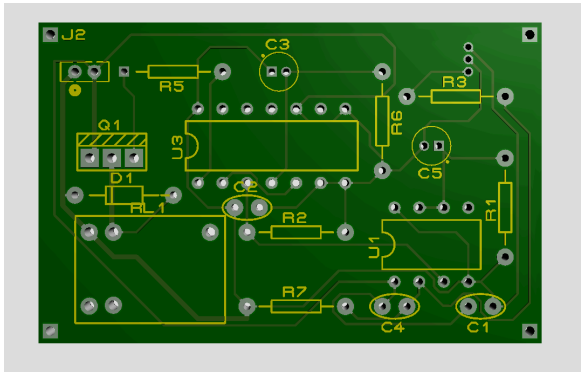
Architecture

block	Role	details
TSOP IR Receiver ↓ Signal Filtering (RC Low-Pass) ↓	Detects infrared signals from the remote contro Removes unwanted noise and smooths the signal	38 kHz IR demodulation added to prevent false triggering caused by noise, which was observed in earlier prototypes.
555 Monostable Pulse Shaper ↓	Generates a fixed-width pulse whenever a valid IR signal is received.	Improved NEC IR protocol signal handling by generating clean trigger pulses
CD4013 Toggle Flip-Flop (D flipflop turned T) ↓	Changes state (ON ↔ OFF) with each pulse from the 555 timer.	added power on reset with an RC circuit to ensure the system starts with “off”
BD139 Transistor Driver ↓ Relay Output Stage	Amplifies the low-current flip-flop output to drive the relay coil Switches the external load (lamp, appliance, etc.) on or off.	respects cd4013 limiting output(Q) source current I implemented a lamp as the load in my breadboard version

Circuit design

1. I implemented a TSOP IR receiver to enable versatile remote control compatibility, allowing the use of most standard TV remotes. A pull-up resistor and a low-pass RC filter (1 kΩ resistor and 100 nF capacitor) were added to improve signal stability and suppress false triggering observed in early prototypes. Through testing and isolation of faults, I identified the TSOP output stage as the source of noise-induced triggering issues.





2.To further process the signal, a 555 timer in monostable mode was used to generate a clean, fixed-width pulse. The pulse width is defined by the RC time constant (1.1RC), set to approximately 500 ms to prevent double triggering while ensuring reliable response timing. Although this introduces a minimum re-trigger delay, it significantly improves system stability.

(proteus isis simulation)

1. For state control, a CD4013 flip-flop was used due to its simplicity, reliability, and low cost, implementing a toggle-based memory function. A power-on reset circuit with a 0.1 s time constant was added to ensure a defined startup state.
2. a BD139 transistor was used as a driver stage to amplify the low-current CMOS output and drive the relay. The flip-flop output sources approximately 0.9 mA, which is sufficient to control the BD139 and energize the relay coil, with a flyback diode included for protection.
3. Finally, the system works on 5 volts, 100 nF caps were added for ics power pins for stability and pullups/pulldowns at CD4013 clock and tsop output.

Debugging & Engineering Challenges

Issue 1: random False triggering

The system initially exhibited unintended switching due to noise on the trigger input. which was later solved with the low pass filter.

Issue 2: ground disturbance by the relay

the relay shifted the ground because it drew a lot of current which messes with the 2 ICs , it was fixed by decoupling caps and large reservoir caps and star ground topology.

Issue 3: random power on state

the system started as “on” sometimes and other times as “off” when first connected to power ,which was fixed by power on reset RC circuit

Results & conclusions

The system reliably toggles a relay using an IR remote input after implementing signal filtering and power stabilization improvements. The final design demonstrates robust hardware-based digital control without microcontroller dependency

QR code for video demonstration